CHAPTER

4

PCR Amplification: Capabilities and Cautions



Forensic science and DNA-typing laboratories have greatly benefited from the discovery of a technique known as the polymerase chain reaction or PCR. First described in 1985 by Kary Mullis and members of the Human Genetics group at the Cetus Corporation (now Roche Molecular Systems), PCR has revolutionized molecular biology through the ability to make hundreds of millions of copies of a specific sequence of DNA in a matter of only a few hours. The impact of PCR has been such that its inventor, Kary Mullis, received the Nobel Prize in Chemistry in 1993—less than 10 years after it was first described.

Without the ability to make copies of DNA molecules, many forensic samples would be impossible to analyze. DNA from crime scenes is often limited in both quantity and quality and obtaining a cleaner, more concentrated sample is normally out of the question (most perpetrators of crimes are, not surprisingly, unwilling to donate more evidence material to aid their prosecution). The PCR DNA amplification technology is well suited to analysis of forensic DNA samples because it is sensitive, rapid, and not as limited by the quality of the DNA as the original restriction fragment length polymorphism (RFLP) methods were when they were used (see Butler 2010, *Fundamentals*, Chapter 3). Some advantages and disadvantages of PCR are summarized in D.N.A. Box 4.1.

PCR BACKGROUND AND BASICS

PCR is an enzymatic process in which a specific region of DNA is replicated over and over again to yield many copies of a particular sequence. This molecular "xeroxing" process involves heating and cooling samples in a precise thermal cycling pattern over ≈30 cycles

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D.N.A. BOX 4.1

ADVANTAGES AND DISADVANTAGES OF PCR

The advantages of PCR amplification for biological evidence include the following:

- Very small amounts of DNA template may be used from as little as a single cell (see Chapter 11).
- DNA degraded to fragments only a few hundred base pairs in length can serve as effective templates for amplification.
- Large numbers of copies of specific DNA sequences can be amplified simultaneously with multiplex PCR reactions.
- Contaminant DNA, such as fungal and bacterial sources, will not amplify because human-specific primers are used.
- Commercial kits are now available for easy PCR reaction setup and amplification.

There are three potential pitfalls that could be considered disadvantages of PCR:

- The target DNA template may not amplify due to the presence of PCR inhibitors in the extracted DNA.
- Amplification may fail due to sequence changes in the primer-binding region of the genomic DNA template.
- Contamination from other human DNA sources besides the forensic evidence at hand or previously amplified DNA samples is possible without careful laboratory technique and validated protocols.

(Figure 4.1). During each cycle, a copy of the target DNA sequence is generated for every molecule containing the target sequence (Figure 4.2). The boundaries of the amplified product are defined by oligonucleotide primers that are complementary to the 3′-ends of the sequence of interest.

In the ideal reaction with 100% amplification efficiency, approximately a billion copies of the target region on the DNA template have been generated after 32 cycles (Table 4.1). However, a reduction in amplification efficiency through PCR inhibition or poor primer annealing leads to lower quantities of PCR product being produced.

The equation for the number of target molecules produced, which incorporates the amplification efficiency, is

$$X_n = X_o(1 + E)^{(N-2)}$$

where X_n is the predicted number of target molecules created, X_o is the number of starting molecules, E is the efficiency of the reaction (between 0% and 100% or 0 to 1), and N is the number of cycles. The N-2 takes into account that for the first two cycles the specific double-stranded target is not yet created. At 100% efficiency, there is a doubling of target molecules with each cycle after the third cycle, where the ends of the PCR product are fully defined by the forward and reverse primers (Table 4.1).

If one of the PCR primers is labeled with a fluorescent dye, then the total number of single-stranded dye-labeled products at the end of each PCR cycle is actually slightly larger

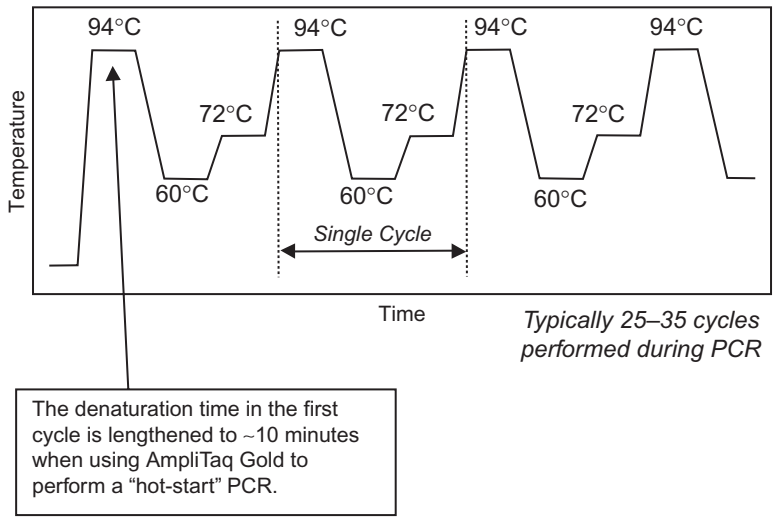


FIGURE 4.1 Thermal cycling temperature profile for PCR. Thermal cycling typically involves 3 different temperatures that are repeated over and over again 25 to 35 times. At 94°C, the DNA strands separate, or 'denature'. At 60°C, primers bind or 'anneal' to the DNA template and target the region to be amplified. At 72°C, the DNA polymerase extends the primers by copying the target region using the deoxynucleotide triphosphate building blocks. The entire PCR process is about 3 hours in duration with each cycle taking ≈5 minutes on conventional thermal cyclers: 1 minute each at 94°C, 60°C, and 72°C and about 2 minutes ramping between the 3 temperatures.

than shown in Table 4.1 because an additional linear amplification occurs after the second cycle off of an extra labeled product.

Following amplification, the PCR product, sometimes referred to as an "amplicon," is then in sufficient quantity that it can be easily measured by a variety of techniques, the most common of which is fluorescence detection (Chapter 6). For this reason, commercial short tandem repeat (STR) typing assays (Chapter 5) label one primer in every primer pair with a fluorescent dye (Figure 4.3). In this way, the amplicon becomes fluorescently labeled and can be detected when the dye label is excited by light of the appropriate wavelength. By using dye labels that can be distinguished from one another in terms of their color (fluorescent emission characteristics), the number of DNA regions that can be analyzed simultaneously is increased. Thus, the primer positions within a sequence and their spacing from one another define the overall PCR product length while the fluorescent dye establishes the amplicon's detection characteristics (e.g., what color if visible dyes are used).

The PCR amplification process, therefore, has a dual purpose: (1) to increase the number of molecules representing a specific target site, and (2) to attach a label, most often a fluorescent dye, that enables detection of the amplicons produced. Both the amplification and labeling elements of the PCR process enhance detection sensitivity and specificity.

PCR is commonly performed with a sample volume in the range of 5 μL to 100 μL. At such low volumes, evaporation can be a problem and accurate pipetting of the reaction components can become a challenge. On the other hand, larger solution volumes lead to thermal equilibrium issues for the reaction mixture because it takes longer for an external

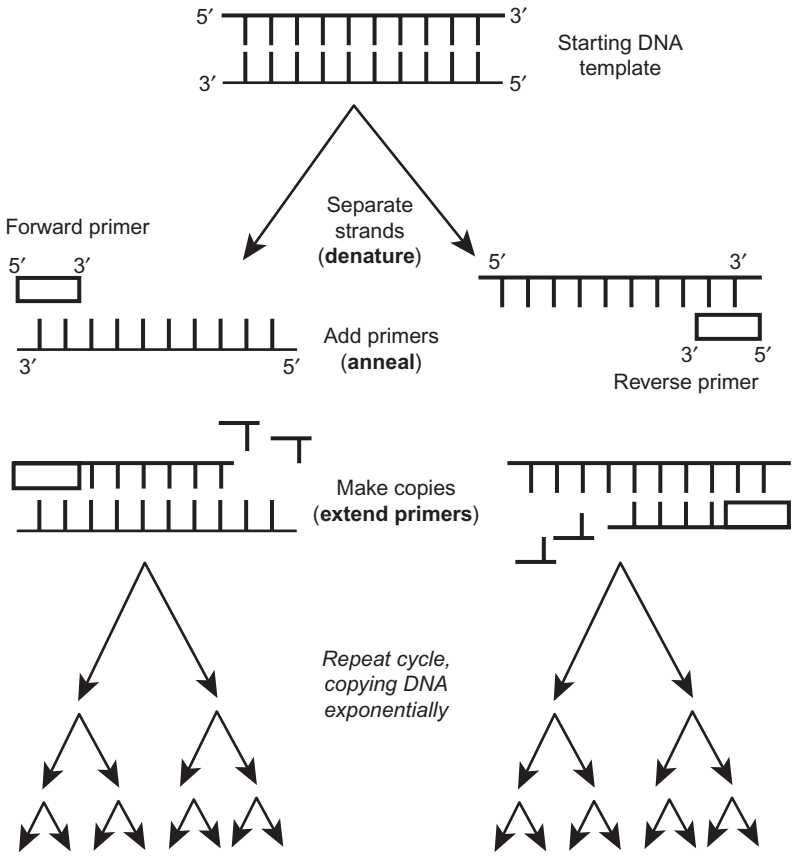


FIGURE 4.2 DNA amplification process with the polymerase chain reaction. In each cycle, the two DNA template strands are first separated (denatured) by heat. The sample is then cooled to an appropriate temperature to bind (anneal) the oligonucleotide primers. Finally the temperature of the sample is raised to the optimal temperature for the DNA polymerase and it extends the primers to produce a copy of each DNA template strand. For each cycle, the number of DNA molecules (with the sequence between the two PCR primers) doubles.

temperature change to be transmitted to the center of a larger solution than a smaller one. Therefore, longer hold times are needed at each temperature, which leads to longer overall thermal cycling times. Most molecular biology protocols for PCR are thus in the 20 μL to 50 μL range.

The sample is pipetted into a variety of reaction tubes designed for use in PCR thermal cyclers. The most common tube in use with 20 μL to 50 μL PCR reactions is a thin-walled 0.2 mL tube. These 0.2 mL tubes can be purchased as individual tubes with and without attached caps or as "strip-tubes" with 8 or 12 tubes connected together in a row. In high-throughput labs, 96-well or 384-well plates are routinely used for PCR amplification. Some work has also been performed with PCR amplification from sample spots on microscope slides that have been overlaid with mineral oil (Schmidt et al. 2006).

TABLE 4.1 Number of Target DNA Molecules Created by PCR with Various Levels of Efficiency. Only at 100% Efficiency Is There a Doubling with Each Cycle. The Sharp Drop-off in the Amount of PCR Product with Lower Efficiencies Shows Why PCR Inhibitors Can Have Such an Impact. The Target DNA Sequence Is Not Completely Defined by the Forward and Reverse Primers Until the Third Cycle.

	Number of Double-Stranded Target Molecules (Specific PCR Product)				
Cycle Number	**100% Efficiency**	**95% Efficiency**	**90% Efficiency**	**50% Efficiency**	**30% Efficiency**
1	0	0	0	0	0
2	0	0	0	0	0
3	2	2	2	2	1
4	4	4	4	2	2
5	8	7	7	3	2
6	16	14	13	5	3
7	32	28	25	8	4
8	64	55	47	11	5
9	128	107	89	17	6
10	256	209	170	26	8
11	512	408	323	38	11
12	1,024	795	613	58	14
13	2,048	1,550	1,165	86	18
14	4,096	3,023	2,213	130	23
15	8,192	5,895	4,205	195	30
16	16,384	11,494	7,990	292	39
17	32,768	22,414	15,181	438	51
18	65,536	43,707	28,844	657	67
19	131,072	85,229	54,804	985	87
20	262,144	166,197	104,127	1,478	112
21	524,288	324,084	197,842	2,217	146
22	1,048,576	631,964	375,900	3,325	190
23	2,097,152	1,232,329	714,209	4,988	247
24	4,194,304	2,403,042	1,356,998	7,482	321
25	8,388,608	4,685,933	2,578,296	11,223	418
26	16,777,216	9,137,569	4,898,763	16,834	543
27	33,554,432	17,818,260	9,307,650	25,251	706
28	67,108,864	34,745,606	17,684,534	37,877	917

(*Continued*)

TABLE 4.1 Number of Target DNA Molecules Created by PCR with Various Levels of Efficiency. Only at 100% Efficiency Is There a Doubling with Each Cycle. The Sharp Drop-off in the Amount of PCR Product with Lower Efficiencies Shows Why PCR Inhibitors Can Have Such an Impact. The Target DNA Sequence Is Not Completely Defined by the Forward and Reverse Primers Until the Third Cycle. (*Continued*)

	Number of Double-Stranded Target Molecules (Specific PCR Product)				
Cycle Number	**100% Efficiency**	**95% Efficiency**	**90% Efficiency**	**50% Efficiency**	**30% Efficiency**
29	134,217,728	67,753,932	33,600,615	56,815	1,193
30	268,435,456	132,120,168	63,841,168	85,223	1,550
31	536,870,912	257,634,328	121,298,220	127,834	2,015
32	1,073,741,824	502,386,940	230,466,618	191,751	2,620
33	2,147,483,648	979,654,533	437,886,574	287,627	3,406
34	4,294,967,296	1,910,326,339	831,984,491	431,440	4,428

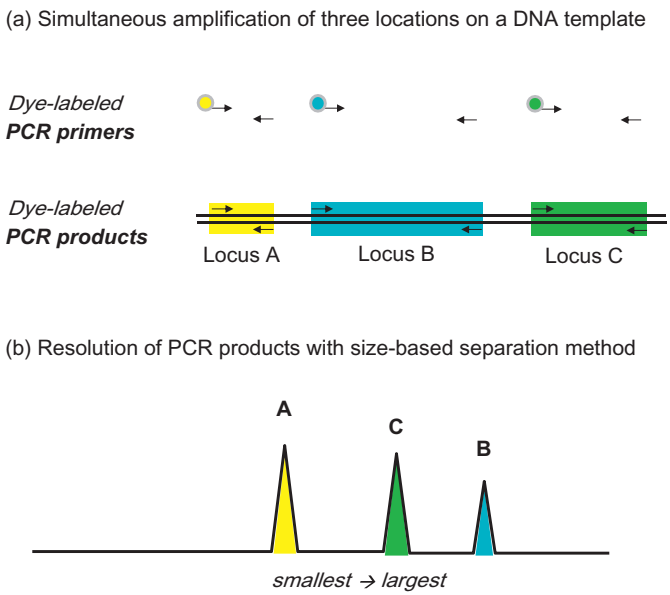


FIGURE 4.3 Schematic of multiplex PCR and how labeled PCR primers create labeled PCR products. (a) Three sets of primers, represented by arrows, are shown here to amplify three different loci on a DNA template. The primers were designed so that the PCR products for locus A, locus B, and locus C would be different sizes and therefore resolvable with a size-based separation system (b).

PCR has been simplified in recent years by the availability of reagent kits that allow a forensic DNA laboratory to simply add a DNA template to a pre-made PCR mix containing all the necessary components for the amplification reaction. These kits are optimized through extensive research efforts on the part of commercial manufacturers. The kits are

typically prepared so that a user adds an aliquot of the kit solution to a particular amount of genomic DNA. The best results with these commercial kits are obtained if the DNA template is added in an amount that corresponds to the concentration range designed for the kit; hence, the need for DNA quantitation and sample quantity normalization described in Chapter 3.

Thermal Cyclers

The instrument that heats and cools a DNA sample in order to perform the PCR reaction is known as a thermal cycler. Accurate and consistent sample heating and cooling is crucial to PCR in order to guarantee consistent results. There are a wide variety of thermal cycler options available from multiple manufacturers. A few of these options are contained in the chapter reference list. These instruments vary in the number of samples that can be handled at a time, the size of the sample tube and volume of reagents that can be handled, and the speed at which the temperature can be changed. Prices for thermal cycling devices range from a few thousand dollars to over 10,000 dollars.

The most prevalent thermal cycler in forensic DNA laboratories today is the GeneAmp 9700 from Applied Biosystems (Foster City, CA). The "9700" is typically used to heat and cool 96 samples in an 8 × 12-well microplate format at a rate of approximately 1°C per second, although its maximum ramp rate is 4°C per second.

Thermal Cycling Parameters

A wide range of PCR cycling protocols have been used for various molecular biology applications. The primary reason that PCR protocols vary is that different primer sequences have different hybridization properties and thus anneal to the DNA template strands at different rates. Annealing time and temperature are some of the most critical parameters to optimize with multiplex PCR. Alternatively, primer sequences may be altered during development to fit a defined PCR thermal cycling protocol.

Most forensic protocols utilize 28 to 30 cycles of PCR with fluorescence detection. To improve detection with low-template DNA samples some laboratories have opted to increase the number of PCR cycles to 31 or even 34 cycles (see Chapter 11). It is important to recognize that increasing sensitivity through additional cycles needs to be carefully validated as additional artifacts may arise.

PCR COMPONENTS & CONTROLS

A PCR sample is prepared by mixing several individual components and then adding deionized water to achieve the desired volume and concentration of each of the components. Commercial kits with pre-mixed components may also be used for PCR. These kits have greatly simplified the use of PCR in forensic DNA laboratories.

The most important components of a PCR sample are the two primers, which are short DNA sequences that precede or "flank" the region to be copied. A primer acts to identify or "target" the portion of the DNA template to be copied. It is a chemically synthesized

TABLE 4.2 Typical Components for PCR Amplification.

Reagent	Optimal Concentration
Tris-HCl, pH 8.3 (25°C)	10–50 mM
Magnesium chloride	1.2–2.5 mM
Potassium chloride	50 mM
Deoxynucleotide triphosphates (dNTPs)	200 μM each dATP, dTTP, dCTP, dGTP
DNA polymerase, thermal stable[a]	0.5–5U
Bovine serum albumin (BSA)	100 μg/mL
Primers	0.1–1.0 μM
Template DNA	1–10 ng genomic DNA

[a]***Taq*** *and* ***Taq****Gold are the two most common thermal stable polymerases used for PCR.*

oligonucleotide that is added in a high concentration relative to the DNA template to drive the PCR sample. Considerable knowledge of the DNA sequence to be copied is required in order to select appropriate primer sequences.

The other components of a PCR sample consist of template DNA that will be copied, deoxynucleotide triphosphate (dNTP) building blocks that supply each of the four nucleotides, and a DNA polymerase that adds the building blocks in the proper order based on the template DNA sequence. The various components and their optimal concentrations are listed in Table 4.2. Thermally stable polymerases that do not fall apart during the near-boiling denaturation temperature steps have been important to the success of PCR. The most commonly used thermally stable polymerase is *Taq*, which comes from a bacterium named *Thermus aquaticus* that inhabits hot springs.

When setting up a set of samples that contain the same primers and reaction components, it is common to prepare a "master mix" that can then be dispensed in equal quantities to each PCR tube. This procedure helps to insure relative homogeneity among samples. Also by setting up a larger number of reactions at once, small pipetting volumes can be avoided, which improves the accuracy of adding each component (and thus the reproducibility of your method). When performing a common test on a number of different samples, the goal should be to examine the variation in the DNA samples not variability in the reaction components used and the sample preparation method.

Controls Used to Monitor PCR

Controls are used to monitor the effectiveness of the chosen experimental conditions and/or the technique of the experimenter. These controls typically include a "negative control," which is the entire PCR reaction mixture without any DNA template. The negative control usually contains water or buffer of the same volume as the DNA template and is useful to assess whether or not any of the PCR components have been contaminated by DNA (e.g.,

you or someone else in your lab). An extraction "blank" is also useful to verify that the reagents used for DNA extraction are free from any extraneous DNA templates.

A "positive control" is a valuable indicator of whether or not any of the PCR components have failed or were not added during the reaction setup phase of experiments conducted. A standard DNA template of known sequence with good-quality DNA should be used for the positive control. This DNA template should be amplified with the same PCR primers as used on the rest of the samples in the batch that is being amplified. The purpose of a positive control is to ensure confidence that the reaction components and thermal cycling parameters are working for amplifying a specific region of DNA.

The challenge with a negative control is that a single PCR tube (or other plasticware that the DNA samples pass through during sample processing) may contain contaminating DNA. If the contaminant DNA is present in only the negative control PCR tube, then the testing analyst may get a false impression that other samples in the batch are contaminated when in fact they are clean. If the contaminant DNA is present in one of the casework DNA samples, then a mixture may result between the contaminant DNA and the original casework sample—and the clean negative control would not be helpful in assessing that a contaminant is present in one of the tested samples.

Peter Gill and Amanda Kirkham from the UK Forensic Science Service (Gill & Kirkham 2004) advocate maintaining a record of profiles that appear in negative controls in order to establish a level of predicted lab contamination from exogenous DNA in contaminated plasticware. An example of a prominent profile from consumable contamination is shown in D.N.A. Box 4.2.

DNA Template Source

Typically following DNA extraction and quantitation, an appropriate amount of genomic DNA would be added to the PCR sample (see Chapter 3). It has been demonstrated that old PCR samples (from DQ alpha or a CTT triplex) contain enough genomic DNA that a subsequent amplification with an STR kit can recover a full STR profile (Patchett et al. 2002). Likewise, STR results have been obtained from old RFLP membranes (Steadman et al. 2008). Hence, storage of old evidence has proven useful when needing to obtain STR typing results for comparison purposes.

Hot-Start PCR

Regular DNA polymerases exhibit some activity below their optimal temperature, which for *Taq* polymerase is 72°C. Thus, primers can anneal non-specifically to the template DNA at room temperature when PCR reactions are being set-up and non-specific products may result. It is also possible at a low temperature for the primers to bind to each other creating products called "primer dimers." These are a particular problem because their small size relative to the PCR products means that they will be preferentially amplified.

Once low-temperature non-specific priming occurs, these undesirable products will be efficiently amplified throughout the remaining PCR cycles. Because the polymerase is busy amplifying these competing products, the target DNA region will be amplified less efficiently. If this happens, you will get less of what you are looking for and you may not have enough specific DNA to run your other tests.

D.N.A. BOX 4.2

CONTAMINANT DNA PROFILE FOUND IN SOME MICROCON DEVICES USED FOR DNA CONCENTRATION

Similar to the German phantom swab contamination discussed in Chapter 1, a number of U.S. forensic DNA laboratories have observed a contaminating DNA profile in Microcon (Millipore, Billerica, MA) centrifugal filter devices used for DNA concentration. Presumably this DNA profile is from an employee of Millipore who works with the Microcon product line as it has appeared in multiple Microcon production lots. The full Identifiler STR profile is included to raise awareness among laboratories who may have observed this full profile or portions of it (due to stochastic PCR amplification from low amounts of DNA) in a reagent blank or PCR negative control.

It is hoped that future lots of Microcon devices will be contaminant-free as more manufacturing control processes are put into place as advocated by leading forensic DNA groups (see D.N.A. Box 4.3).

Locus	DNA Type
amelogenin	X,X
D8S1179	13,15
D21S11	29,30
D7S820	10,10
CSF1PO	10,11
D3S1358	16,17
TH01	7,9
D13S317	8,9
D16S539	13,13
D2S1338	20,25
D19S433	14,14
vWA	17,18
TPOX	8,11
D18S51	14,16
D5S818	11,11
FGA	20,23

Low-temperature mispriming can be avoided by initiating PCR at an elevated temperature, a process usually referred to as "hot-start" PCR. Hot-start PCR may be performed by introducing a critical reaction component, such as the polymerase, after the temperature of the sample has been raised above the desired annealing temperature (e.g., 60°C). This minimizes the possibility of mispriming and misextension events by not having the polymerase present during reaction setup. However, this approach is cumbersome and time-consuming when working with large numbers of samples. Perhaps a more important disadvantage is the fact that the sample tubes must be opened at the thermal cycler to introduce the essential component, which gives rise to a greater opportunity for cross-contamination between samples. As will be discussed in the next section, a modified form of *Taq* DNA polymerase has been developed that requires thermal activation and thus enables a closed-tube hot-start

D.N.A. BOX 4.3

ENFSI-SWGDAM-BSAG STATEMENT TO MANUFACTURERS

In the July 2010 issue of *Forensic Science International: Genetics*, the European Network of Forensic Science Institutes (ENFSI), the FBI Laboratory's Scientific Working Group on DNA Analysis Methods (SWGDAM) composed of U.S. and Canadian members, and the Australian Biology Specialist Advisory Group (BSAG) published a joint statement to commercial manufacturers of disposable plasticware and other reagents used by forensic DNA laboratories worldwide (Gill et al. 2010). Recognizing that consumable products have been observed with detectable DNA in the past (see Chapter 1 for discussion on the "German phantom" from contaminated buccal swabs), these groups proposed that a new product grade be made available for forensic applications that would include:

(a) Automation of manufacturing lines
(b) Minimizing interaction of staff with manufacturing lines
(c) Ensuring that products which come into contact with staff are adequately protected (i.e., staff gowned and masked, with feet and hair covered)
(d) Use of positive pressure airflow through HEPA filters, i.e., Class 10,000 clean room standard or equivalent
(e) Continual quality control (QC) checks, using PCR analysis, of a number of consumable items—preferably sensitive enough to detect a single cell with all profiles observed being recorded for future comparison
(f) Once the consumables have been manufactured, an additional stage may be used to physically destroy any DNA contaminant that may have been inadvertently introduced.
(g) QC checks and use of process controls to ensure the post-production treatment(s) have been effective.

The ENFSI-SWGDAM-BSAG joint position statement also advocates that forensic DNA laboratories maintain an elimination database for screening DNA results as appropriate (Gill et al. 2010). The elimination database should contain (1) forensic laboratory staff (including DNA laboratory, evidence recovery, and other support staff DNA profiles along with former staff members), (2) regular visitors to laboratory areas, (3) contractors such as service engineers and cleaning staff that have access to the laboratory, (4) manufacturing staff of forensic kits and consumables (where possible), (5) unexplained profiles observed in negative controls (see D.N.A. Box 4.2), and (6) police personnel, forensic medical examiners, mortuary staff and pathologists who would potentially have contact with biological samples provided to the forensic DNA laboratory.

Regarding the manufacturing staff profiles, the statement goes on to propose that DNA profiles are either held anonymously on a central database or held by individual suppliers with access available to their customers (Gill et al. 2010). Part of the concern within the United States is that the Genetic Information Nondiscrimination Act of 2008 (GINA) prevents employers from retaining genetic information on their employees (see http://www.genome.gov/24519851). There is an exception for forensic DNA laboratories in creating staff elimination databases

D.N.A. BOX 4.3 (*cont'd*)

for detecting potential sample contamination but it is unclear how this clause impacts manufacturers of reagents used in forensic DNA testing. The forensic DNA exception clause within GINA, Section 202(b)(6), states: "where the employer conducts DNA analysis for law enforcement purposes as a forensic laboratory or for purposes of human remains identification, and requests or requires genetic information of such employer's employees, but only to the extent that such genetic information is used for analysis of DNA identification markers for quality control to detect sample contamination."

Promega Corporation (Madison, WI), which supplies STR kits and other reagents for human identity testing, has responded that they meet or exceed all of the ENFSI-SWGDAM-BSAG recommendations to reduce the chance of human-to-product contamination (Pearson et al. 2010). It is hoped that other manufacturers will respond in a similar fashion.

Sources:

Gill, P., et al. (2010). Manufacturer contamination of disposable plastic-ware and other reagents—an agreed position statement by ENFSI, SWGDAM and BSAG. Forensic Science International: Genetics, *4, 269–270.*

Pearson, K., et al. (2010). Manufacturing high-quality forensic tools—Promega quality standards. Profiles in DNA, 13(2). *Available at http://www.promega.com/profiles/1302/1302_08.html.*

PCR. This enzyme, named AmpliTaq Gold, has greatly benefited the specificity of PCR amplifications.

AmpliTaq Gold DNA Polymerase

AmpliTaq Gold DNA polymerase is a chemically modified enzyme that is rendered inactive until heated. An extended pre-incubation of 95°C, usually for 10 or 11 minutes, is used to activate the AmpliTaq Gold. The chemical modification involves a derivatization of the epsilon-amino groups of the lysine residues. At a pH below 7.0 the chemical modification moieties fall off and the activity of the polymerase is restored (Birch et al. 1998).

The pH of the Tris buffer in the PCR reaction varies with temperature; higher temperatures cause the solution pH to go down by approximately 0.02 pH units with every 1°C. A Tris buffer with pH 8.3 at 25°C will go down to pH ≈6.9 at 95°C. Thus, not only is the template DNA well denatured but the polymerase is activated just when it is needed, and not in a situation where primer dimers and mispriming can occur as easily.

Other DNA polymerases have been developed recently that are rendered active for hot-start PCR with shorter times (e.g., 1 to 2 minutes instead of 10 to 11 minutes). These enzymes can help reduce the overall time for PCR amplification and enable rapid PCR efforts.

MULTIPLEX PCR

The polymerase chain reaction permits more than one region to be copied simultaneously by simply adding more than one primer set to the reaction mixture (Edwards & Gibbs 1994). The simultaneous amplification of two or more regions of DNA is commonly known as multiplexing or multiplex PCR (Figure 4.3).

For a multiplex reaction to work properly the primer pairs need to be compatible. In other words, (1) the primer annealing temperatures should be similar and (2) excessive regions of complementarity between the primers should be avoided to prevent the formation of primer dimers that will cause the primers to bind to one another instead of the template DNA. The addition of each new primer in a multiplex PCR reaction exponentially increases the complexity of possible primer interactions (Butler et al. 2001).

Each new PCR application is likely to require some degree of optimization in either the reagent components or thermal cycling conditions. Multiplex PCR is no exception. In fact, multiplex PCR optimization is more of a challenge than singleplex reactions because so many primer-annealing events must occur simultaneously without interfering with each other. Extensive optimization is normally required to obtain a good balance between the amplicons of the various loci being amplified (Kimpton et al. 1996, Markoulatos et al. 2002).

The variables that are examined when trying to obtain optimal results for a multiplex PCR amplification include concentrations for many of the reagents listed in Table 4.2 as well as the thermal cycling temperature profile. Primer sequences and concentrations along with magnesium concentrations are usually the most crucial for multiplex PCR. Extension times during thermal cycling are often increased for multiplex reactions in order to give the polymerase time to fully copy all of the DNA targets. Obtaining successful co-amplification with well-balanced PCR product yields sometimes requires redesign of primers and tedious experiments with adjusting primer concentrations.

Primer design for the STR DNA markers discussed in Chapter 5 includes some additional challenges. Primers need to be adjusted on the STR markers to achieve good size separation between loci labeled with the same fluorescent dye. In addition, the primers must produce robust amplifications with good peak height balance between loci as well as specific amplification with no non-specific products that might interfere with proper interpretation of a sample's DNA profile. Finally, primers should produce a maximal non-template-dependent "A" addition to all PCR products to ease data interpretation.

PCR Primer Design

Well-designed primers are probably the most important components of a good PCR reaction. The target region on the DNA template is defined by the position of the primers. PCR yield is directly affected by the annealing characteristics of the primers. For the PCR to work efficiently, the two primers must be specific to the target region, possess similar annealing temperatures, not interact significantly with each other or themselves to form "primer dimers," and be structurally compatible. Likewise, the sequence region to which the primers bind must be fairly well conserved because if the sequence changes from one DNA template to the next then the primers will not bind appropriately. Some general guidelines to optimal PCR primer design are listed in Table 4.3 (see also Dieffenbach et al. 1993, Butler 2005).

TABLE 4.3 General Guidelines for PCR Primer Design.

Parameter	Optimal Values
Primer length	18–30 bases
Primer T_m (melting temperature)	55–72°C
Percentage GC content	40–60%
No self-complementarity (hairpin structure)	≤3 contiguous bases
No complementarity to other primer (primer dimer)	≤3 contiguous bases (especially at the 3′-ends)
Distance between two primers on target sequence	<2000 bases apart
Unique oligonucleotide sequence	Best match in BLAST[a] search
T_m difference between forward and reverse primers in pair	≤5°C
No long runs with the same base	<4 contiguous bases

[a]*BLAST search examines similarity of the primer to other known sequences that may result in multiple binding sites for the primer and thus reduce the efficiency of the PCR amplification reaction. BLAST searches may be conducted via the Internet: http://www.ncbi.nlm.nih.gov/BLAST.*

A number of primer design software packages are commercially available including Primer Express (Applied Biosystems, Foster City, CA) and Oligo (Rychlik & Rhoads 1989; Molecular Biology Insights, Cascade, CO). These programs use thermodynamic "nearest neighbor" calculations to predict annealing temperatures and primer interactions with themselves or other possible primers (Mitsuhashi 1996, SantaLucia 1998).

The Internet has become a valuable resource for tools that aid primer selection. For example, a primer design program called Primer 3 (Rozen & Skaletsky 2000) is available on the World Wide Web through the Whitehead Institute (http://frodo.wi.mit.edu/). With Primer 3, the user inputs a DNA sequence and specifies the target region within that sequence to be amplified. Parameters such as PCR product size, primer length, and desired annealing temperature may also be specified by the user. The program then ranks the best PCR primer pairs and passes them back to the user over the Internet. Primer 3 works well for quickly designing singleplex primer pairs that amplify just one region of DNA at a time. The National Center of Biotechnology Information recently released a combined Primer-BLAST tool that enables finding specific PCR primers through combining the features of Primer3 and BLAST (see http://www.ncbi.nlm.nih.gov).

Multiplex PCR Optimization

The development of an efficient multiplex PCR reaction requires careful planning and numerous tests and efforts in the area of primer design and balancing reaction components (Shuber et al. 1995, Henegariu et al. 1997, Markoulatos et al. 2002). A range of thermal cycling parameters including annealing temperatures and extension times are often examined in developing the final protocol.

Obtaining a nicely balanced multiplex PCR reaction with each PCR product having a similar yield is a challenging task. With the widespread availability of commercial kits,

individual forensic laboratories rarely perform PCR optimization experiments anymore. Rather, internal validation studies focus on performance of the multiplex with varying conditions around the optimal parameters supplied with the kit protocol. For example, PCR product yields in the form of STR peak heights produced by a commercial kit might be evaluated at the optimal annealing temperature (e.g., 59°C) and 2°C and 4°C higher and lower (e.g., 55°C, 57°C, 61°C, and 63°C). Differences, if any, would then be noted relative to the optimal annealing temperature supplied in the kit protocol.

Primer concentrations are one of the largest factors in a multiplex PCR reaction determining the overall yield of each amplicon (Schoske et al. 2003). Repeated experiments and primer titrations are usually performed to achieve an optimal balance. Concentrations of magnesium chloride and dNTPs are typically increased slightly relative to singleplex reactions. A thorough evaluation of performance for a multiplex will also involve addition and removal of primer sets to see if overall balance in other amplification targets is affected.

The availability of 5-dye detection systems has enabled development of multiplexes capable of amplifying and analyzing in excess of 20 short tandem repeat loci (Butler et al. 2002, Hanson & Ballantyne 2004, Hill et al. 2009) and 52 single nucleotide polymorphism markers (Sanchez et al. 2006).

The basic principles of STR assay development have been described in several publications from the Human Identity Project Team at the National Institute of Standards and Technology (NIST) (Schoske et al. 2003, Butler 2005, Hill et al. 2009). NIST also provides a downloadable software program called *AutoDimer* to check hairpin structures (intra-primer comparisons) and primer dimers (inter-primer comparisons) when developing multiplex assays (Vallone & Butler 2004). A recent software tool called *Multiplex Manager* enables PCR primer and assay design (Holleley & Geerts 2009; http://www.multiplexmanager.com/).

PCR INHIBITION

As mentioned in Chapter 2, the PCR amplification process can be affected by substances known as "inhibitors," which interfere or prevent the DNA amplification process from occurring properly (Bessetti 2007). These inhibitors can be present in DNA samples collected from crime scenes. Outdoor crimes may leave body fluid such as blood and semen on soil, sand, wood, or leaf litter that contain substances which may co-extract with the perpetrator's DNA and prevent PCR amplification. Textile dyes, leather, and wood surfaces from interior crime scenes may also contain DNA polymerase inhibitors.

Inhibitors can (1) interfere with the cell lysis necessary for DNA extraction, (2) interfere by nucleic acid degradation or capture, and (3) inhibit polymerase activity thus preventing enzymatic amplification of the target DNA (Wilson 1997). Occasionally substances such as textile dyes from clothing or hemoglobin from red blood cells can remain with the DNA throughout the sample preparation process and interfere with the polymerase to prevent successful PCR amplification (Akane et al. 1994, DeFanchis et al. 1988, Rådström et al. 2004).

The result of amplifying a DNA sample containing an inhibitor, such as hematin, is a loss of the alleles from the larger-sized STR loci or even complete failure of all loci. Some examples of PCR inhibitors and their sources are found in Table 4.4.

TABLE 4.4 Some PCR Inhibitors and Their Sources.

PCR Inhibitor	Possible Source
Heme (hematin)	Blood
Melanin	Tissue and hair
Polysaccharides	Feces, plant material
Bile salts	Feces
Humic compounds	Soil
Urea	Urine
Indigo dye (denim)	Clothing (blue jeans)
Collagen	Tissue
Myoglobin	Muscle tissue
Calcium ions	Bone

Samples containing PCR inhibitors often produce partial profile results that look similar to a degraded DNA sample (see Chapter 10). Thus, failure to amplify the larger STR loci for a sample can be either due to degraded DNA where there are not enough intact copies of the DNA template or due to the presence of a sufficient level of PCR inhibitor that reduces the activity of the polymerase. Reduced-size STR amplicons can aid in recovery of information from a sample that is inhibited since smaller PCR products may be amplified more efficiently than larger ones.

Solutions to PCR Inhibition

A nice review of strategies to generate PCR-compatible samples has been published (Rådström et al. 2004, see also Wilson 1997). PCR inhibitors may be removed or their effects reduced by one or more of the following solutions. The genomic DNA template may be diluted, which also dilutes the PCR inhibitor, and re-amplified in the presence of less inhibitor. Alternatively, more DNA polymerase can be added to overcome the inhibitor. With this approach some fraction of the *Taq* polymerase binds to the inhibiting molecule(s) and removes them from the reaction so that the rest of the *Taq* can do its job and amplify the DNA template. In addition, polymerases other than *Taq* have been shown to work well with blood and feces, which typically inhibit PCR when performed with *Taq* DNA polymerase (Al-Soud & Rådström 1998). A recent effort mixed multiple DNA polymerases to improve PCR performance in the presence of various types of inhibitors (Hedman et al. 2010).

Additives to the PCR reaction, such as bovine serum albumin (BSA) (Comey et al. 1994, Kreader 1996) or betaine (Al-Soud & Rådström 2000), have been shown to prevent or minimize the inhibition of PCR. For this reason, BSA is a common ingredient in most STR typing kits. It should be noted though that BSA quality can vary between sources and that acetylated BSA is actually inhibitory to PCR.

A sodium hydroxide treatment of DNA has also been shown to neutralize inhibitors of *Taq* polymerase (Bourke et al. 1999). The addition of aluminum ammonium sulfate proved helpful to prevent the co-purification of inhibitors with DNA from soil samples (Braid et al. 2003). Finally, a separation step may be performed prior to PCR to separate the extracted DNA from the inhibiting compound. Centricon-100 and Microcon-100 filters have been used for this purpose (Comey et al. 1994) as have low-melt agarose gel plugs (Moreira 1998).

Where possible, sample collection should avoid adding inhibitors. For example, swabbing of surfaces containing inhibitors such as denim instead of processing cuttings of sample can help. Ideally, as noted in Chapter 2, the process of DNA extraction purifies the desired DNA away from inhibitors. If not, then monitoring an internal positive control (IPC) in qPCR assays used for DNA quantitation can at least make an analyst aware that PCR inhibitors are present (Kontanis & Reed 2006).

MODIFICATIONS TO IMPROVE PCR

A number of modifications to PCR buffer, primers, and DNA polymerases have enabled improvements in PCR performance in recent years. These modifications have also enabled direct PCR amplification from bloodstains as well as more rapid PCR amplification as will be discussed later in the chapter.

PCR Primer Improvements

Another source of potential improvement in PCR reactions is the oligonucleotide primer itself. Figure 4.4 illustrates some of the potential modifications that have been made in PCR primers used in STR typing assays. These modifications include use of fluorescent dye tags (including energy transfer primers), non-nucleotide linkers to alter the electrophoretic mobility of the PCR product, extra forward or reverse primers to cover multiple primer binding site possibilities, and high-stability analogs such as locked nucleic acids (Table 4.5).

Work with Other DNA Polymerases

In an effort to improve success with forensic DNA samples, alternative polymerases have been explored. Johannes Hedman and colleagues at the Swedish National Laboratory of Forensic Science investigated Bio-X-Act Short (Bioline, London, UK), Ex*Taq* Hot Start (Takara Bio Inc., Shiga, Japan), KAPA2G Robust (KAPA Biosystems, Cape Town, South Africa), Omni*Taq* (DNA Polymerase Technologies, St. Louis, MO, USA), PicoMaxx High Fidelity (Stratagene, La Jolla, CA, USA), r*Tth* (Applied Biosystems), *Taq*, and *Tth* (Roche Diagnostics, Mannheim, Germany) compared to AmpliTaq Gold (Applied Biosystems) (Hedman et al. 2009). They found that use of Bio-X-Act Short, Ex*Taq* Hot Start, or PicoMaxx High Fidelity provided significant improvement in amplification success from AmpliTaq Gold-inhibited PCR samples.

Furthermore, a blend of *ExTaq* Hot Start and PicoMaxx High Fidelity enabled 34 of 42 "inhibited" crime scene samples to yield better DNA profiles than could be obtained with standard

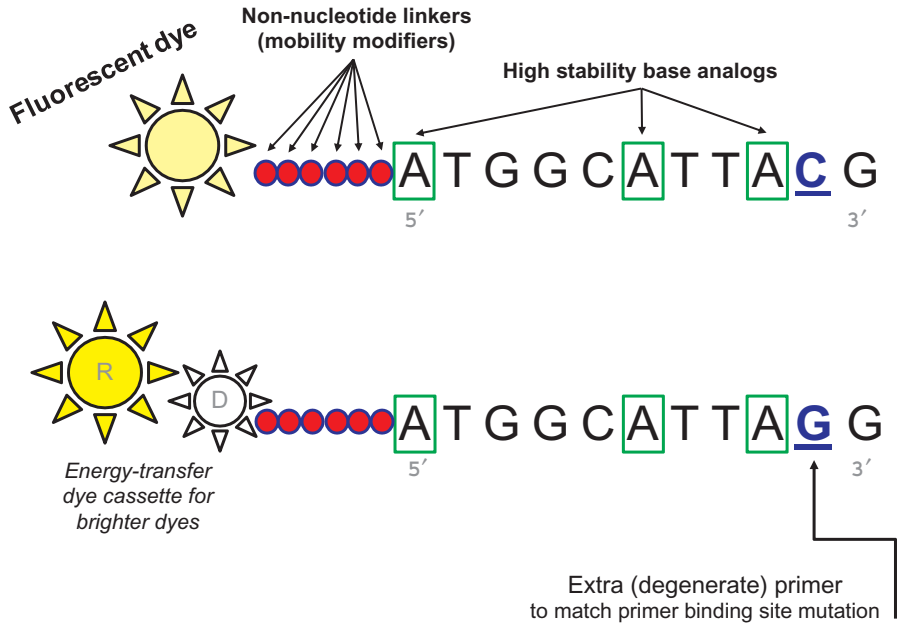


FIGURE 4.4 Representation of a portion of a PCR primer sequence with potential modifications (for more information on the purpose of each modification, see Table 4.5).

TABLE 4.5 PCR Primer (Oligonucleotide) Labels and Possible Modifications as Illustrated in Figure 4.4.

Modification	Purpose
Fluorescent dye tag	Enables laser-induced fluorescence detection of single-stranded DNA and with multiple dyes, spectral resolution enables detection of PCR products that possess overlapping size ranges
Mobility modifiers (non-nucleotide linkers)	Enables an apparent size shift during electrophoresis such that STR allele size ranges may be altered while retaining original primer sequences to avoid "discovering" primer binding site mutations
High stability nucleic acid analogs	Increases primer annealing stability to aid with DNA amplification in the presence of potential PCR inhibitors
Degenerate (extra) primer	Provides for amplification of a DNA sample containing a primer binding site mutation that would normally result in the primer not annealing and allele dropout occurring
Location along the target DNA sequence	Alters the overall PCR product size—if closer to the STR repeat region, then a "miniSTR" amplicons may improve amplification success with degraded or inhibited DNA samples—if either closer or further away from the STR repeat region, then amplicon size ranges may fit more optimally into another STR kit or "in-house" assay

AmpliTaq Gold amplifications (Hedman et al. 2010). Polymerase blends have also successfully aided rapid PCR efforts (Vallone et al. 2008).

STOCHASTIC EFFECTS

Forensic DNA specimens often possess low levels of DNA. When amplifying very low levels of DNA template, a phenomenon known as stochastic fluctuation can occur. Stochastic effects, which are an unequal sampling of the two alleles present from a heterozygous individual, result when only a few DNA molecules are used to initiate PCR. Under conditions of limited template, the PCR primers used to amplify a specific region may not consistently find and hybridize to the entire set of DNA molecules present in the PCR amplification reaction. With a heterozygous locus, where two alleles are present, the unequal sampling of the alleles can result in failure to detect one or both of the alleles. Loss of a single allele is referred to as "allele drop-out" while loss of both alleles is termed "locus drop-out."

PCR reactions involving DNA template levels below approximately 100 pg of DNA, or about 15 diploid copies of genomic DNA, have been shown to exhibit allele dropout. False homozygosity results if one of the alleles fails to be detected (Figure 4.5).

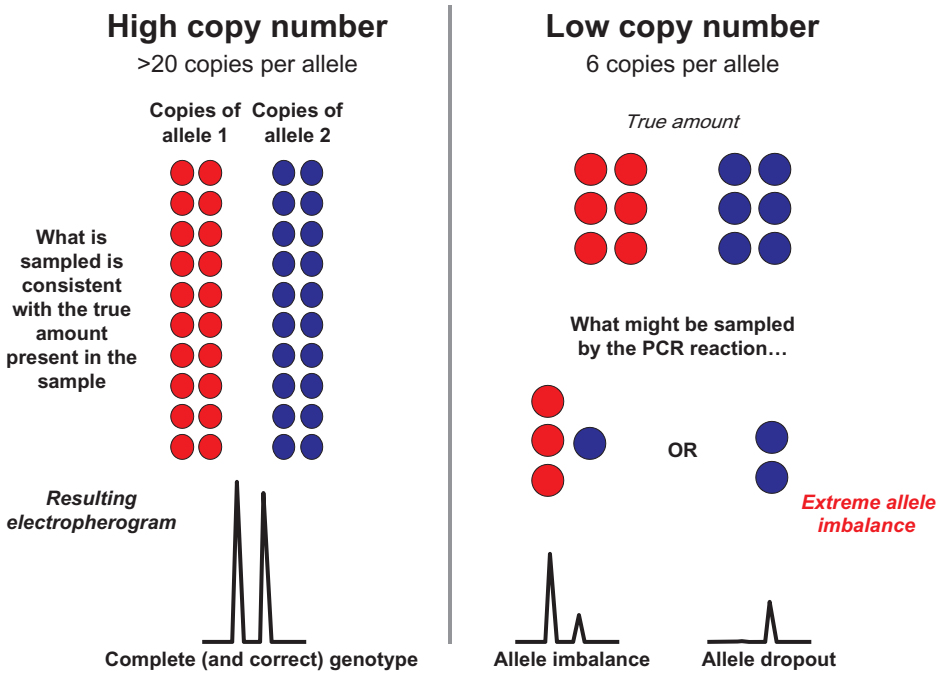


FIGURE 4.5 An illustration of stochastic statistical sampling that may occur with low template amounts where the two alleles sampled at a genetic locus during PCR are represented as red and blue circles. Allele imbalance, allele dropout, or even locus dropout may occur resulting in an electropherogram that does not accurately reflect the original DNA sample.

Stochastic (random) variation is a fundamental physical law of the PCR amplification process when examining low amounts of DNA (Butler & Hill 2010). Stochastic effects are manifest as a fluctuation of results between replicate analyses. In other words, PCR amplifying the same DNA extract twice can result in different alleles being detected at a locus.

Stochastic artifacts can be avoided by adjusting the cycle number of the PCR reaction such that approximately 20 or more copies of target DNA are required to yield a successful typing result (Walsh et al. 1992). However, efforts have been made to obtain results with low-copy number (LCN) or low template DNA (LT-DNA) testing (Gill et al. 2000). The challenges of LCN work and trying to interpret data obtained with less than 100 pg of DNA template will be addressed in Chapter 11.

Since stochastic effects cannot be avoided when testing small quantities of DNA, there are essentially two schools of thought on how to handle these types of samples: (1) stop testing or interpreting data before you go low enough to be in the stochastic realm, or (2) try to limit the impact of the stochastic variation by additional testing and careful interpretation guidelines based on validation studies (Butler & Hill 2010).

Those who advocate the second approach usually enhance their method sensitivity, such as increasing the number of PCR cycles, in order to get as much out of the limited sample as possible. The "enhanced interrogation" approach typically involves replicate testing and the development of consensus profiles (see Chapter 11).

Some scientists have insisted that whole genome amplification (WGA) techniques can be used to increase the number of target molecules and limit or reduce stochastic effects when working with low-level DNA. Unfortunately, the early cycles of WGA are also subject to stochastic effects and therefore an uneven allelic representation may be created with WGA. Chapter 11 covers more on this topic.

PRECAUTIONS AGAINST CONTAMINATION

The sensitivity of PCR necessitates constant vigilance on the part of the laboratory staff to ensure that contamination does not affect DNA typing results. Contamination of PCR reactions is always a concern because the technique is very sensitive to low amounts of DNA. A scientist setting up the PCR reaction can inadvertently add his or her own DNA to the reaction if he or she is not careful. Likewise, the police officer or crime scene technician collecting the evidence can contaminate the sample if proper care is not taken. For this reason, each piece of evidence should be collected with clean tweezers or handled with disposable gloves that are changed frequently.

To aid discovery of laboratory contamination, everyone in a forensic DNA laboratory is typically genotyped in order to have a record of possible contaminating DNA profiles. This is often referred to as a staff elimination database. Laboratory personnel should be appropriately gowned during interactions with samples prior to PCR amplification (Rutty et al. 2003). The appropriate covering includes lab coats and gloves as well as facial masks and hairnets to prevent skin cells or hair from falling into the amplification tubes. These precautions are especially critical when working with miniscule amounts of sample or sample that has been degraded.

Some tips for avoiding contamination with PCR reactions in a laboratory setting include:

- Pre- and post-PCR sample processing areas should be physically separated. Usually a separate room or a containment cabinet is used for setting up the PCR amplification reactions.
- Equipment, such as pipettors, and reagents for setting up PCR should be kept separate from other laboratory supplies, especially those used for analysis of PCR products.
- Disposable gloves should be worn and changed frequently.
- Reactions may also be set up in a laminar flow hood, if available.
- Aerosol-resistant pipette tips should be used and changed on every new sample to prevent cross-contamination during liquid transfers. Pipette tips should never be reused. Even a tiny droplet of PCR product left in a pipette tip may contain millions of copies of the amplifiable sequence. By comparison, a nanogram of human genomic DNA contains only about 300 copies of single-copy DNA markers (see D.N.A. Box 3.1).
- Reagents should be carefully prepared to avoid the presence of any contaminating DNA or nucleases.
- Ultraviolet irradiation of laboratory PCR setup space when the area is not in use and cleaning workspaces and instruments with isopropanol and/or 10% bleach solutions help ensure that extraneous DNA molecules are destroyed prior to DNA extraction or PCR setup (Kwok & Higuchi 1989, Prince & Andrus 1992).

PCR product carryover results from amplified DNA contaminating a sample that has not yet been amplified. Because the amplified DNA is many times more concentrated than the unamplified DNA template, it will be preferentially copied during PCR and the unamplified sample will be masked. The inadvertent transfer of even a very small volume of a completed PCR amplification to an unamplified DNA sample can result in the amplification and detection of the "contaminating" sequence. For this reason, the evidence samples are typically processed through a forensic DNA laboratory prior to the suspect reference samples to avoid any possibility of contaminating the evidence with the suspect's amplified DNA.

Reagent and Consumable Contamination

As was noted in Chapter 1, it is important that collection swabs be DNA-free prior to sample collection. Likewise, other DNA extraction reagents and consumables such as tubes and tips need to be clean prior to use so that the DNA profile being processed accurately reflects the individual collected and not a consumable contaminant.

Unfortunately with a highly sensitive technique like PCR, contamination does occur in some instances. The DNA results from a contaminant DNA profile observed in multiple U.S. forensic DNA laboratories is shown in D.N.A. Box 4.2. In order to significantly reduce the potential for contaminant DNA profiles being put on national DNA databases, efforts need to be taken to have DNA-free products. Recommendations from several prominent forensic DNA organizations regarding improved manufacturing protocols to produce DNA-free products are listed in D.N.A. Box 4.3.

Consumable Sterilization Efforts

Research has found that ethylene oxide treatment, which is routinely used in hospital equipment sterilization, is an effective method to remove background DNA (Archer et al. 2010, Shaw et al. 2008). In fact, dual ethylene oxide treatment effectively removed DNA molecules such that profiles were not obtained from items intentionally contaminated with 10 µL of blood or saliva while gamma irradiation and electron beam methods of sterilization were not as successful at removing the contaminating DNA (Archer et al. 2010). Unfortunately, ethylene oxide is a highly toxic gas and so consumables and reagents to be used in a forensic DNA laboratory would need to either come pretreated with ethylene oxide by the original manufacturer or be sent out to a treatment facility (and risk contamination with the additional handling).

Another study found that treating consumables with a 2 h autoclave treatment could eliminate nanogram quantities of DNA (Gefrides et al. 2010). Still others prefer ultraviolet irradiation to eliminate exogenous DNA from plasticware and water used in preparing reagents (Tamariz et al. 2006).

NEW CAPABILITIES WITH IMPROVEMENTS IN PCR

Improvements in various components and aspects of the PCR process have led to some new capabilities including direct PCR, which enables bypassing DNA extraction and quantitation, and rapid PCR, which permits obtaining results in a fraction of the time traditionally needed to perform multiplex PCR amplification.

Direct PCR

Amplification directly from a blood stain can save DNA extraction time and reagents and may enable greater sensitivity from small samples as the wash steps involved in DNA extraction often reduce the overall extraction efficiency and sample recovery.

The creation of new DNA polymerases (Wang et al. 2004), such as the Phusion High-Fidelity DNA polymerase (Finnzymes Espoo, Finland), enables increased affinity for double-stranded DNA to help overcome PCR inhibitors, incorporation of more nucleotides per binding event, and a decrease in the number of binding events for DNA strand elongation (see http://www.finnzymes.com/directpcr/).

Next-generation STR kits, such as Identifiler Direct and PowerPlex 16 HS, have been engineered to enable PCR amplification in the presence of heme and other PCR inhibitors so that biological samples may be directly amplified without prior purification (Wang et al. 2009, Chang et al. 2010).

Rapid PCR

The current forensic DNA typing process takes about 8 to 10 hours. The longest single step in this process is multiplex PCR amplification, which requires approximately three hours using manufacturer-supplied protocols for commercial STR typing kits. There is great

interest in developing portable, rapid DNA-typing devices for a number of applications. For example, the ability to perform multiplex PCR amplification with commonly used STR markers in a few minutes rather than hours could open new potential biometric applications for DNA testing including analysis of individuals while they wait at a country border or an airport.

STR kit manufacturers have validated their kits with 1°C/s temperature ramp rates and dwell times of around one minute per temperature step. The time required is for 28 to 32 cycles is typically two-and-a-half to three hours. Part of this time is a 10 minute front-end hot-start to active the AmpliTaq Gold DNA polymerase. A 30 to 60 minute 60°C soak is used at the end of thermal cycling to enable full adenylation of the PCR products produced.

Using different DNA polymerases and a faster temperature ramp rate with shorter dwell times at each temperature, PCR amplifications of STR typing kits have been reduced to approximately 35 minutes on a conventional GeneAmp 9700 thermal cycler, which can change temperatures at a maximum rate of 4°C/s (Vallone et al. 2008). With faster ramp rate cyclers, STR typing results have been obtained in as low as 15 minutes (Giese et al. 2009, Vallone et al. 2009).

The improved capabilities in terms of both direct PCR and rapid PCR will open new avenues and applications for DNA testing. Only time will tell how far and fast we can go in terms of PCR amplification.

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